SafeKeep Support Agent

Unit-Test Plan

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Contents

1. Introduction....3
2. Test approach....3
3. Environment and execution....4
4. Test cases....5
5. Scenario validation with the real model....14
6. Traceability matrix....15
7. Coverage summary and exit criteria....16
8. Not covered by unit tests....17
9. Revision history....17

1. Introduction

1.1 Purpose

This plan defines the unit tests for the SafeKeep Support Agent. Every test case is derived from the Software Functional Specification and traces to at least one requirement ID. The traceability matrix in Section 6 shows that every requirement is covered.

1.2 Scope

Table 1. Scope

In scope (unit and slice tests)	Out of scope here (validated in Phase 4)
Event intake, validation, deduplication, concurrency	Visual rendering of UI screens
Investigation state machine and resolution rules	Quality of the real LLM's reasoning (see Section 5)
Agent loop with a scripted fake LLM	Live HTTP calls to Ollama (one manual smoke test)
Response validation guardrails	Performance and load
Tool gateway, permissions, approval gate, verification	
Configuration, security rules, history, health, metrics	
Simulated application adapters and seed-data rules	

2. Test approach

2.1 Test levels

Table 2. Test levels

Level	What it is	Used for
Unit	Plain JUnit 5 + Mockito/AssertJ. One class under test, collaborators replaced by fakes or mocks. No Spring context.	Almost all logic: validator, gateway, state machine, approval, orchestrator.
Slice	Spring Boot test slices (@WebMvcTest) that load only the web and security layer.	HTTP status codes, role rules, CSRF, API-key filter.

2.2 Test doubles

These doubles make the tests fast, repeatable and independent of the network.

Table 3. Test doubles

Double	Replaces	Behavior
FakeLlmClient	OllamaLlmClient	Returns a scripted sequence of responses (tool calls, final answers, errors) and records every request, so tests can assert what the model was sent.
Fake adapters	BackupServerAdapter, StorageServerAdapter	Return canned tool results; can be told to throw, delay or change state.

Double	Replaces	Behavior
TestApplication	A third application	Test-only YAML + adapter used to prove a new application needs no core change.
T-ERR seed data	Failing application	Test-only scenario whose get_logs tool and purge action throw, to test error handling.
Mocked RunScheduler	Background worker	Tests start runs explicitly, step by step, instead of on a background thread.
Fixed Clock	System time	Deterministic timestamps, expiry and metrics.
Synchronous executor	Worker thread pool	Runs investigations on the test thread unless a test needs real concurrency.
In-memory repositories / H2	Database	Isolated data per test.

2.3 Conventions

- **ID format:** TC-<AREA>-<nn>, e.g. TC-APR-07. The ID appears in the test method's @DisplayName so reports map directly to this plan.
- **Structure:** every test case states Given (setup), When (steps) and Then (expected result and assertions).
- **Priority:** P1 = safety or core correctness, must pass before any demo. P2 = important behavior. P3 = edge case.
- **Parameterized tests:** one test case may run several input rows (JUnit @ParameterizedTest); each row is reported separately.

3. Environment and execution

Table 4. Environment

Item	Value
Runtime	Java 21, Maven 3.9+
Frameworks	JUnit 5, Mockito, AssertJ, Spring Boot Test, JaCoCo
External services	None. Ollama is not needed for unit tests.
Run all tests	`./mvnw test`
Run one area	`mvn test -Dtest=ApprovalServiceTest`
Coverage report	`mvn verify` then open target/site/jacoco/index.html

Example of how a test case from this plan looks in code:

```
@Test
@DisplayName("TC-VAL-02 CONFIRMED finding without evidence is downgraded")
void confirmedFindingWithoutEvidenceIsDowngraded() {
    // Given
    var ctx = investigationWithEvidence(ok("E1"), ok("E2"), error("E3"));
    var output = finalOutput(finding(CONFIRMED, "Heap usage is 93%", List.of()));
    // When
    var result = validator.validate(output, ctx);
    // Then
    var f = result.findings().get(0);
    assertThat(f.kind()).isEqualTo(SUSPECTED);
    assertThat(f.downgraded()).isTrue();
}
```

4. Test cases

4.1 Event intake

Test class: EventIntakeTest (Spring + in-memory H2)

Common setup: backup-server and storage-server test configs; API keys bk-test and st-test; fixed clock 2026-11-07T10:00Z; run scheduler mocked.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-EVT-01 P1	No investigations exist	POST a valid HIGH_MEMORY event for backup-server-01 with key bk-test	202 CREATED. One investigation in RECEIVED, eventCount 1, correlationKey backup-server|HIGH_MEMORY|backup-server-01. One EventRecord. Run scheduled once.	FR-EVT-01 FR-EVT-02 FR-INV-01
TC-EVT-02 P1	Valid event body	POST with (a) no key, (b) st-test for a backup-server event, (c) a random key	401 for each row. No investigation. Intake failure recorded with reason AUTH.	FR-EVT-01 FR-EVT-05 NFR-SEC-01
TC-EVT-03 P1	Event body without resource	POST with key bk-test	400. Error lists field resource. No investigation. Intake failure recorded.	FR-EVT-02 FR-EVT-05
TC-EVT-04 P1	Application billing-app is not configured	POST event for billing-app	422 UNKNOWN_APPLICATION. Intake failure recorded.	FR-EVT-01 FR-EVT-05
TC-EVT-05 P2	storage-server configured	POST eventType DISK_FULL	422 UNKNOWN_EVENT_TYPE. hint lists the configured event types.	FR-EVT-01 NFR-USAB-01
TC-EVT-06 P3	Clock at 10:00Z	POST with timestamp 25 h in the future	400. No investigation.	FR-EVT-02

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-EVT-07 P1	INV-1 created from evt-1	POST evt-1 again	200 DUPLICATE_IGNORED with INV-1. eventCount stays 1. No new EventRecord. No new run.	FR-EVT-04
TC-EVT-08 P1	INV-1 is INVESTIGATING	POST evt-2 with the same correlation key	200 LINKED with INV-1. eventCount 2. Two EventRecords. No new investigation or run.	FR-EVT-04
TC-EVT-09 P2	INV-1 is WAITING_FOR_INFO	POST evt-2 with the same key	200 LINKED (WAITING_FOR_INFO counts as open).	FR-EVT-04
TC-EVT-10 P1	INV-1 is COMPLETED	POST evt-3 with the same key	202 CREATED INV-2 with recurrenceOfId INV-1. INV-1 unchanged.	FR-EVT-04
TC-EVT-11 P2	INV-1 open for backup-server| HIGH_MEMORY| backup-server-01	POST (a) same type for backup-server-02, (b) HIGH_CPU for backup-server-01	A new investigation for each row. No recurrence link.	FR-EVT-04
TC-EVT-12 P1	No investigations	10 threads POST 10 different eventIds with the same key at once (CountDownLatch)	Exactly 1 investigation. eventCount 10. 10 EventRecords.	FR-EVT-04 NFR-SCAL-01
TC-EVT-13 P1	LlmClient and adapters are mocks	POST a valid event	No interaction with LlmClient or any adapter during the request. Response returned before the run starts.	FR-EVT-03 NFR-RESP-01
TC-EVT-14 P2	Key bk-test	POST a rejected event whose header and body contain bk-test	Stored intake failure and captured log output do not contain bk-test.	NFR-SEC-01

4.2 Investigation state machine

Test class: InvestigationStateMachineTest, InvestigationTest, AsyncRunSchedulerTest (unit); InvestigationLifecycleTest (Spring)

Common setup: Investigation in the given state; progress publisher and audit service mocked.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-INV-01 P1	Each allowed transition from the transition table (8 rows)	Apply the transition	State updated. Audit STATE_CHANGED with from and to.	FR-INV-02

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-INV-02 P1	Disallowed rows: RECEIVED→COMPLETED, RECEIVED→WAITING_FOR_INFO, WAITING_FOR_INFO→COMPLETED, COMPLETED→WAITING_FOR_INFO, UNABLE_TO_COMPLETE→COMPLETED	Apply the transition	IllegalStateTransitionException. State unchanged. Audit INVALID_TRANSITION.	FR-INV-02
TC-INV-03 P2	Progress subscriber registered	Run through RECEIVED→INVESTIGATING→COMPLETED	One progress event per transition with investigationId, from and to.	FR-INV-03
TC-INV-04 P1	INVESTIGATING with evidence E1	Run fails with reason “AI service unavailable” (implemented together with TC-AGT-05)	UNABLE_TO_COMPLETE, stateReason set, agent message posted, E1 kept.	FR-INV-04
TC-INV-05 P2	Two INVESTIGATING and one COMPLETED in the database	Startup recovery runs	The two become UNABLE_TO_COMPLETE (“Interrupted by restart”). COMPLETED unchanged.	FR-INV-04 NFR-REL-01
TC-INV-06 P2	New investigation	Create, then move to COMPLETED	Resolution starts UNKNOWN and is not changed by state transitions.	FR-INV-02
TC-INV-07 P2	Worker pool with 1 thread and no queue; the thread is busy	Submit a second run, free the thread, run the retry	The second run is kept as pending, then executed on retry. Nothing is lost.	NFR-REL-01 FR-EVT-03

4.3 Agent orchestrator

Test class: AgentOrchestratorTest (Spring + FakeLlmClient)

Common setup: FakeLlmClient with a scripted response list; fake adapters with canned S-1A data; synchronous executor; budget 8 tool calls.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-AGT-01 P1	Script: get_metrics, get_top_processes, get_job_sessions, valid final answer	Run the investigation	ToolCalls E1–E3 with status OK in order. Findings persisted. COMPLETED. firstRecommendationAt set.	FR-AI-01 FR-DATA-01 FR-DATA-03 MET-01
TC-AGT-02 P1	backup-server HIGH_MEMORY event	Inspect the first request sent to the LLM	Tools are exactly the enabled HIGH_MEMORY read subset. No ACTION tools. No storage-server tools. Actions appear only in the “proposable actions” text.	FR-AI-02 FR-DATA-02

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-AGT-03 P1	Admin disabled get_logs	Run the investigation	get_logs is absent from every tool list sent to the LLM.	FR-ACT-01 FR-DATA-02
TC-AGT-04 P1	Script requests tools indefinitely	Run the investigation	Exactly 8 tool executions, then a partial-result request. missingInformation not empty. COMPLETED.	FR-AI-04
TC-AGT-05 P1	One successful tool call, then LlmClient always throws	Run the investigation	3 attempts on the failing call (1 + 2 retries). UNABLE_TO_COMPLETE with “AI service unavailable”. E1 kept.	NFR-REL-01 FR-INV-04
TC-AGT-06 P2	Final answer malformed once, then valid	Run the investigation	One repair request containing the parse error. COMPLETED.	NFR-REL-01
TC-AGT-07 P2	Final answer malformed twice	Run the investigation	UNABLE_TO_COMPLETE with “Invalid AI output”.	NFR-REL-01
TC-AGT-08 P1	Adapter throws for get_logs	Run the investigation	ToolCall status ERROR. Next LLM request contains a tool message with ADAPTER_ERROR. Run continues to COMPLETED.	NFR-REL-01
TC-AGT-09 P1	Final answer has questionForUser while unused tools remain	Run the investigation	The model is sent back once to use its tools (“Do not ask the user yet”); on the second question: WAITING_FOR_INFO and an AGENT_QUESTION message.	FR-DATA-04
TC-AGT-10 P1	Investigation WAITING_FOR_INFO	User replies	INVESTIGATING, then COMPLETED. The reply is in the next LLM request.	FR-USR-03 FR-DATA-04
TC-AGT-11 P1	Investigation COMPLETED	User asks “Will this come back?”	Exactly one LLM call, with no tools offered. Answer message saved. Still COMPLETED. No new ToolCalls.	FR-USR-04
TC-AGT-12 P1	COMPLETED with a PROPOSED action	User adds context (row a) or requests more investigation (row b)	INVESTIGATING. Prior findings (and context) in the request. Proposal EXPIRED. Budget reset to 8.	FR-USR-03 FR-USR-05
TC-AGT-13 P2	A tool returns 5,000 characters	Run the investigation	Content sent to the model is at most 1,500 characters and ends with a truncation note.	FR-AI-01
TC-AGT-14 P1	Application config contains an API key	Capture every LLM request during an S-1A run	No request contains the key value or an apiKey field.	NFR-SEC-01
TC-AGT-15 P1	Two investigations with different scripts	Run both concurrently on two threads	Each has only its own ToolCalls and findings. Evidence numbering independent (both start at E1).	NFR-SCAL-01
TC-AGT-16 P2	Script repeats the identical call get_job_sessions()	Run the investigation	Executed once only; the model is told “Already called with the same arguments; see E1”.	FR-AI-01

4.4 Response validator

Test class: ResponseValidatorTest (unit)

Common setup: Investigation with evidence E1 (OK), E2 (OK), E3 (ERROR); backup-server config.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-VAL-01 P1	CONFIRMED finding citing E1	Validate	Stays CONFIRMED. downgraded = false.	FR-AI-03
TC-VAL-02 P1	CONFIRMED finding citing nothing	Validate	Becomes SUSPECTED. downgraded = true.	FR-AI-03
TC-VAL-03 P1	CONFIRMED finding citing only E3 (ERROR)	Validate	Becomes SUSPECTED. downgraded = true.	FR-AI-03
TC-VAL-04 P2	Finding cites E1 and E9	Validate	evidenceIds = [E1].	FR-DATA-03
TC-VAL-05 P1	proposedAction: (a) unknown name, (b) disabled action, (c) a storage-server action	Validate	No proposal kept. A MANUAL recommendation with the action's configured instructions (generic text for unknown) is added.	FR-ACT-06
TC-VAL-06 P1	No findings and empty missingInformation	Validate	missingInformation = [“Cause could not be determined”]. Resolution UNKNOWN.	FR-AI-04
TC-VAL-07 P1	Model returns resolution RESOLVED (row a) or PARTIALLY_RESOLVED (row b)	Validate	Resolution replaced by UNKNOWN; only the system sets these values.	FR-ACT-05
TC-VAL-08 P2	findings missing, or wrong types	Validate	Invalid result with a readable error message for the repair prompt.	NFR-REL-01
TC-VAL-09 P2	questionForUser present	Validate	Next state = WAITING_FOR_INFO.	FR-DATA-04
TC-VAL-10 P1	Enabled action proposed, resolution UNKNOWN	Validate	Proposal dropped; a note explains the problem is not shown to be present (rule R-V8).	FR-ACT-03

4.5 Tool gateway

Test class: ToolGatewayTest (unit)

Common setup: backup-server config; fake adapter that records invocations; tool timeout 5 s.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-GW-01 P1	Tool get_passwords is not defined	Agent requests it	UNKNOWN_TOOL. Adapter not invoked. ToolCall status DENIED.	FR-ACT-02
TC-GW-02 P1	get_logs disabled by admin	Agent requests it	TOOL_DISABLED. Adapter not invoked.	FR-DATA-02 FR-ACT-01

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-GW-03 P1	purge_stale_sessions is an ACTION	Agent loop requests it as a tool	ACTION_NOT_CALLABLE. Adapter not invoked.	FR-ACT-03
TC-GW-04 P2	get_metrics(hours: 1–168, default 24)	Request with (a) hours = "24" (string), (b) hours missing	Adapter receives integer 24 in both rows.	FR-ACT-02
TC-GW-05 P1	Same tool	Request with hours "0", 500, "abc", metric "disk", or an unknown argument server	INVALID_ARGUMENT naming the field and allowed values. Adapter not invoked.	FR-ACT-02
TC-GW-06 P2	Adapter sleeps 6 s	Agent requests the tool	TIMEOUT after about 5 s. ToolCall status ERROR.	NFR-REL-01
TC-GW-07 P2	Adapter throws an exception whose message contains a key	Agent requests the tool	ADAPTER_ERROR with a sanitized message that does not contain the key.	NFR-REL-01 NFR-SEC-01
TC-GW-08 P1	One OK, one DENIED, one ERROR call	Execute all three	Exactly one ToolCall and one AuditEntry per call, with tool, arguments, status and duration.	FR-ACT-07

4.6 Approval gate

Test class: ApprovalTest (Spring + real seed data), ConditionEvaluatorTest (unit)

Common setup: S-1A adapter with mutable state; proposal purge_stale_sessions (LOW, SUPPORT_ENGINEER); fixed clock; users alice, sam, adam.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-APR-01 P1	PROPOSED purge_stale_sessions	sam approves	APPROVED → EXECUTING → SUCCEEDED. Verify tool executed. Resolution RESOLVED. Success message in conversation.	FR-ACT-03 FR-ACT-05
TC-APR-02 P1	Rows: alice/purge (deny), sam/purge (allow), sam/run_expiration_cleanup (deny), adam/run_expiration_cleanup (allow), alice/retry_report_job (allow)	User approves	Allowed rows execute. Denied rows: AccessDeniedException, status stays PROPOSED, adapter not invoked, audit APPROVAL_DENIED.	FR-ACT-03 NFR-SEC-02
TC-APR-03 P1	Proposal already approved and executed	sam approves again	Rejected (status not PROPOSED). Adapter invoked once in total.	FR-ACT-04
TC-APR-04 P1	Proposal created at 10:00	Clock moves to 10:16, sam approves	Approval fails. Status EXPIRED. Not executed.	FR-ACT-03
TC-APR-05 P1	Stored params modified after proposal (hash mismatch)	sam approves	Not executed. Integrity failure audited.	FR-ACT-04

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-APR-06 P1	Proposal with params {jobId: Daily-Gold-VMware}	sam approves	Adapter receives exactly the stored params.	FR-ACT-04
TC-APR-07 P1	S-2A: retry_failed_vms; storage still full	sam approves	Adapter returns OK but verification shows job FAILED. Status FAILED. Resolution UNRESOLVED. Message states the failure and reason.	FR-ACT-05
TC-APR-08 P1	Adapter throws during the action	sam approves	Status FAILED. Resolution UNRESOLVED. Never SUCCEEDED.	FR-ACT-05 NFR-REL-01
TC-APR-09 P2	S-1C: cleanup brings pool to 92% (threshold 90%)	adam approves	SUCCEEDED. Resolution PARTIALLY_RESOLVED.	FR-ACT-05
TC-APR-10 P1	PROPOSED action	sam rejects with reason “change window closed”	REJECTED, decidedBy sam, reason audited. Not executed.	FR-ACT-07
TC-APR-11 P2	Admin disables the action after it was proposed	sam approves	Not executed (TOOL_DISABLED). Manual instructions added to recommendations.	FR-ACT-06 FR-ACT-01
TC-APR-12 P3	Proposal ID belongs to another investigation	sam approves via the wrong investigation	Not found. Not executed.	NFR-SEC-02
TC-APR-13 P2	Verification conditions (9 rows: thresholds, strings, &&, nested fields, missing fields)	Evaluate each condition against a result	Each row returns the expected true/false.	FR-ACT-05

4.7 Configuration

Test class: ApplicationRegistryTest (unit), SecurityTest (MockMvc), NewApplicationTest (Spring)

Common setup: Production YAML files from src/main/resources; test YAML files from src/test/resources.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-CFG-01 P1	Both application YAML files	Load the registry	2 applications with their event types, tool kinds, per-event-type tool subsets, actions with risk and requiredRole.	FR-CFG-01
TC-CFG-02 P2	Rows: (a) action without requiredRole, (b) action without verify	Load the registry	(a) defaults to SUPPORT_ENGINEER. (b) startup fails with a message naming the application and action.	FR-CFG-01
TC-CFG-03 P1	get_logs enabled	adam disables it via PATCH /api/applications/…	Disabled in the registry; audit CONFIG_CHANGED with actor adam. (sam gets 403, covered by TC-SEC-02.)	FR-ACT-01 NFR-SEC-02

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-CFG-04 P1	test-app YAML and TestAdapter in test resources only	Send an event, run with the fake LLM, approve its action	Full flow works. No production class is modified.	FR-CFG-04 NFR-MAINT-01
TC-CFG-05 P2	Environment variable for storage-server API key missing	Build the registry	storage-server marked unavailable with an error message; empty key matches nothing.	FR-CFG-03 NFR-SEC-01
TC-CFG-06 P2	Registry with a secret key	Print the application (toString)	The key does not appear.	NFR-SEC-01

4.8 Security

Test class: SecurityTest (MockMvc), HealthAndMetricsTest (unit)

Common setup: Demo users alice, sam, adam; MockMvc.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-SEC-01 P1	No session	GET /investigations as a browser (Accept: text/html); GET /api/investigations	Page redirects to /login. API returns 401.	NFR-SEC-02
TC-SEC-02 P1	Rows: /api/health as alice (403) and sam (200); PATCH /api/applications/… as sam (403) and adam (200)	Call each endpoint	Status codes as listed.	NFR-SEC-02 FR-CFG-03
TC-SEC-03 P2	sam logged in	POST approve without a CSRF token	403. Not executed.	NFR-SEC-02
TC-SEC-04 P2	Masking helper	Mask a message containing bk-test-key	Output contains **** instead of the key.	NFR-SEC-01
TC-SEC-05 P1	No session, no CSRF token	POST /api/events with a valid key (row a) and a wrong key (row b)	(a) 202 Accepted. (b) 401.	FR-EVT-01 NFR-SEC-01

4.9 User interaction and history

Test class: UserInteractionTest (Spring)

Common setup: Investigations INV-1 and INV-2 with findings, evidence, a proposal and messages.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-USR-01 P1	INV-1 with mixed findings	Load the detail view model	Summary, state, resolution, eventCount, recurrence link, findings grouped by kind with evidence references and downgraded flag, ordered recommendations, proposal.	FR-USR-01
TC-USR-02 P1	Subscriber on INV-1	Progress events for INV-1 and INV-2 are published	Subscriber receives only INV-1 events.	FR-USR-02 FR-INV-03
TC-USR-03 P2	sam gives feedback useful = true	sam then gives useful = false	One feedback record for sam, value false.	FR-USR-06
TC-USR-04 P2	Investigations across apps, states and dates	Filter state COMPLETED, app storage-server, date range	Exactly the matching investigations, newest first.	FR-CFG-02

4.10 Health and metrics

Test class: HealthAndMetricsTest (unit), ApprovalTest (MET-03)

Common setup: Mocked Ollama client; fixture investigations with fixed timestamps.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-HLT-01 P2	Rows: Ollama unreachable; reachable but model missing	Compute health	LLM status DOWN with error; DEGRADED with “model not pulled”.	FR-CFG-03
TC-HLT-02 P3	60 intake failures	Load health	Latest 50, newest first.	FR-EVT-05 FR-CFG-03
TC-MET-01 P2	First recommendation after 10 s, 20 s, 40 s	Compute MET-01	Median = 20 s.	MET-01
TC-MET-02 P2	4 investigations, 3 with evidence-backed findings; 3 feedback, 2 positive	Compute MET-02	75% with evidence; 67% positive.	MET-02
TC-MET-03 P2	4 executed actions, 3 SUCCEEDED	Compute MET-03	Success rate 75%. Executions without approval = 0.	MET-03

4.11 Simulated applications and seed data

Test class: SeedDataAdapterTest (unit)

Common setup: Production seed data loaded per scenario; reset before each test.

ID	Given (setup)	When (steps)	Then (expected result and assertions)	Req.
TC-ADP-01 P1	Each core scenario (S-1A, 1B, 1C, 2A, 2C, 3A, 3B, 4A)	Call the scenario's tools	Raw values match fixture spot checks (e.g. S-1A get_job_sessions: 1,180 total, 8 running).	UC-1 UC-2 UC-3 FR-DATA-01
TC-ADP-02 P1	All seed JSON files	Scan for keys rootCause, cause, diagnosis, explanation	No such keys exist (seed data holds raw observations only).	FR-AI-03
TC-ADP-03 P1	S-1A loaded	Execute purge_stale_sessions, then read metrics	Memory 93% → 61%. Stale sessions 0.	UC-1 FR-ACT-05
TC-ADP-04 P1	S-2A loaded	Execute retry_failed_vms, then get_job_details	Job still FAILED with “insufficient space”.	UC-2 FR-ACT-05
TC-ADP-05 P1	S-3A loaded	Call get_dependency_status	vcenter-prod reported unreachable with an expired certificate.	UC-3
TC-ADP-06 P2	S-1A after purge	Reset the scenario	State identical to initial load.	UC-1
TC-ADP-07 P2	S-1A loaded	get_logs(contains=session-cleanup, level=WARN); search_runbook(stale sessions memory)	Only matching log lines returned; runbook search returns excerpts.	FR-DATA-01

5. Scenario validation with the real model

Unit tests prove the plumbing and guardrails with a scripted model. They cannot prove that the real model reasons well. These checks run in Phase 4 against llama3.2:3b and are recorded in the validation summary.

Table 5. Scenario validation checklist

ID	Scenario	Pass criteria
SV-1A	High memory, stale sessions	Stale sessions reported as CONFIRMED; stuck cleanup as SUSPECTED; purge proposed; RESOLVED after approval.
SV-1B	High CPU, auth loop	Login rate and session count cited; token reuse SUSPECTED; recurrence warning given.
SV-1C	Storage pool 98%	Expiration gap identified; cleanup requires ADMIN; PARTIALLY_RESOLVED after approval.
SV-2A	Backup job failed, storage full	Insufficient space cited; retry FAILED reported honestly; manual steps given.
SV-2C	Backup job failed, no clear cause	No cause stated as CONFIRMED; missing information listed; question asked.
SV-3A	vCenter certificate expired	Affected and unaffected operations listed; no action; escalation steps.
SV-3B	Storage server unreachable	Queued backups and unavailable restores listed.
SV-4A	Unexpected restart	Disk I/O errors and kernel panic cited; hardware escalation; no action.

Because the model is non-deterministic, each scenario runs three times; a scenario passes if all three runs meet the criteria, and any failure is logged with the prompt refinement that fixed it.

6. Traceability matrix

This matrix maps every requirement to the test cases that verify it, based on the Req. column in Section 4. Every requirement is covered by at least one test case. Use cases are additionally verified by the scenario checks in Section 5.

Table 6. Requirement-to-test traceability

Requirement	Test cases	Count
FR-EVT-01	TC-EVT-01, TC-EVT-02, TC-EVT-04, TC-EVT-05, TC-SEC-05	5
FR-EVT-02	TC-EVT-01, TC-EVT-03, TC-EVT-06	3
FR-EVT-03	TC-EVT-13, TC-INV-07	2
FR-EVT-04	TC-EVT-07, TC-EVT-08, TC-EVT-09, TC-EVT-10, TC-EVT-11, TC-EVT-12	6
FR-EVT-05	TC-EVT-02, TC-EVT-03, TC-EVT-04, TC-HLT-02	4
FR-INV-01	TC-EVT-01	1
FR-INV-02	TC-INV-01, TC-INV-02, TC-INV-06	3
FR-INV-03	TC-INV-03, TC-USR-02	2
FR-INV-04	TC-INV-04, TC-INV-05, TC-AGT-05	3
FR-DATA-01	TC-AGT-01, TC-ADP-01, TC-ADP-07	3
FR-DATA-02	TC-AGT-02, TC-AGT-03, TC-GW-02	3
FR-DATA-03	TC-AGT-01, TC-VAL-04	2
FR-DATA-04	TC-AGT-09, TC-AGT-10, TC-VAL-09	3
FR-AI-01	TC-AGT-01, TC-AGT-13, TC-AGT-16	3
FR-AI-02	TC-AGT-02	1
FR-AI-03	TC-VAL-01, TC-VAL-02, TC-VAL-03, TC-ADP-02	4
FR-AI-04	TC-AGT-04, TC-VAL-06	2
UC-1	TC-ADP-01, TC-ADP-03, TC-ADP-06 + SV-1A, SV-1B, SV-1C	3
UC-2	TC-ADP-01, TC-ADP-04 + SV-2A, SV-2C	2
UC-3	TC-ADP-01, TC-ADP-05 + SV-3A, SV-3B	2
FR-USR-01	TC-USR-01	1
FR-USR-02	TC-USR-02	1
FR-USR-03	TC-AGT-10, TC-AGT-12	2
FR-USR-04	TC-AGT-11	1
FR-USR-05	TC-AGT-12	1
FR-USR-06	TC-USR-03	1
FR-ACT-01	TC-AGT-03, TC-GW-02, TC-APR-11, TC-CFG-03	4
FR-ACT-02	TC-GW-01, TC-GW-04, TC-GW-05	3
FR-ACT-03	TC-VAL-10, TC-GW-03, TC-APR-01, TC-APR-02, TC-APR-04	5
FR-ACT-04	TC-APR-03, TC-APR-05, TC-APR-06	3

Requirement	Test cases	Count
FR-ACT-05	TC-VAL-07, TC-APR-01, TC-APR-07, TC-APR-08, TC-APR-09, TC-APR-13, TC-ADP-03, TC-ADP-04	8
FR-ACT-06	TC-VAL-05, TC-APR-11	2
FR-ACT-07	TC-GW-08, TC-APR-10	2
FR-CFG-01	TC-CFG-01, TC-CFG-02	2
FR-CFG-02	TC-USR-04	1
FR-CFG-03	TC-CFG-05, TC-SEC-02, TC-HLT-01, TC-HLT-02	4
FR-CFG-04	TC-CFG-04	1
NFR-SEC-01	TC-EVT-02, TC-EVT-14, TC-AGT-14, TC-GW-07, TC-CFG-05, TC-CFG-06, TC-SEC-04, TC-SEC-05	8
NFR-SEC-02	TC-APR-02, TC-APR-12, TC-CFG-03, TC-SEC-01, TC-SEC-02, TC-SEC-03	6
NFR-REL-01	TC-INV-05, TC-INV-07, TC-AGT-05, TC-AGT-06, TC-AGT-07, TC-AGT-08, TC-VAL-08, TC-GW-06, TC-GW-07, TC-APR-08	10
NFR-RESP-01	TC-EVT-13	1
NFR-SCAL-01	TC-EVT-12, TC-AGT-15	2
NFR-MAINT-01	TC-CFG-04	1
NFR-USAB-01	TC-EVT-05	1
MET-01	TC-AGT-01, TC-MET-01	2
MET-02	TC-MET-02	1
MET-03	TC-MET-03	1

7. Coverage summary and exit criteria

Table 7. Test cases per area

Area	Test cases	P1	P2	P3
Event intake	14	9	4	1
Investigation state machine	7	3	4	0
Agent orchestrator	16	12	4	0
Response validator	10	7	3	0
Tool gateway	8	5	3	0
Approval gate	13	9	3	1
Configuration	6	3	3	0
Security	5	3	2	0
User interaction and history	4	2	2	0
Health and metrics	5	0	4	1
Simulated applications and seed data	7	5	2	0
Total	**95**	**58**	**34**	**3**

Table 8. Exit criteria

Criterion	Target
P1 test cases	100% pass before any demo

Criterion	Target
P2 / P3 test cases	Pass, or failure documented with reason in the validation summary
Requirement coverage	Every requirement in Section 6 has at least one passing test
Line coverage (JaCoCo)	At least 80% for intake, investigation, agent, tools and approval packages
Scenario validation	All 8 core scenarios pass Section 5 criteria

7.1 Execution result

Table 9. Execution result

Item	Result
Command	`./mvnw clean test`
Tests executed (including parameterized rows)	145
Failures / errors / skipped	0 / 0 / 0
Duration	About 14 seconds
Failures found and fixed during development	TC-SEC-01: an unauthenticated request without an Accept header received 401 instead of a redirect. The application was correct (browsers send Accept: text/html); the test was changed to act like a browser.

8. Not covered by unit tests

Table 10. Validated outside unit tests

Area	Why	How it is validated
UI rendering and layout	Visual quality is not meaningful to assert in unit tests	Demo walkthrough in Phase 4
Real model reasoning	Non-deterministic	Scenario validation (Section 5)
Ollama HTTP integration	Needs a running Ollama server	OllamaLlmClientIT, tagged ollama, excluded from mvn test and run manually
Performance and load	Out of scope for the demo	Not tested; noted in the spec's out-of-scope section

9. Revision history

Table 11. Revision history

Revision	Changes
1	Initial plan, derived from the Functional Spec before implementation (88 test cases).
2	Updated after implementation: actual test class names; new TC-INV-07, TC-AGT-16, TC-VAL-10, TC-APR-13, TC-CFG-06, TC-SEC-05, TC-ADP-07; TC-AGT-09 now covers the ask-before-tools guard; TC-ADP-05 and TC-CFG-03 revised; TC-SEC-01 uses a browser Accept header; execution result added (Section 7.1).